

Program 1 - Multiply two 16 bit binary numbers

```
AREA MULTIPLY , CODE, READONLY
ENTRY

START
MOV R1, #6400
MOV R2, #3200
MUL R3, R1, R2

STOP B STOP
END
```

Program 2 - Find the sum of first 10 integer numbers

```
AREA SUM, CODE, READONLY
ENTRY

MOV R1, #10
MOV R2, #0

LOOP
ADD R2, R2, R1
SUBS R1, #0X01
BNE LOOP

STOP B STOP
END
```

Program 3 - Find factorial of a number

```
AREA FACTORIAL , CODE, READONLY
ENTRY

START
MOV R0, #4
MOV R1, R0

FACT SUBS R1, R1, #1
CMP R1, #1
BEQ STOP
MUL R3, R0, R1
MOV R0, R3
BNE FACT

STOP
STOP B STOP
END
```

Program 4 - Add an array of 16 bit numbers and store the 32 bit result in internal RAM

```
AREA ADDITION , CODE, READONLY
ENTRY

START
LOOP
MOV R5, #6
MOV R0, #0
LDR R1, =VALUE1
LDR R2, [R1], #2
LDR R3, MASK
AND R2, R2, R3
```

```
ADD R0,R0,R2
SUBS R5,R5,#1
CMP R5,#0
BNE LOOP
```

```
LDR R4,=RESULT
STR R0,[R4]
```

```
XSS B XSS
```

```
MASK DCD 0X0000FFFF
VALUE1 DCW 0X1111,0X2222,0X3333,0XAAAA,0BBBBB,0CCCC
```

```
AREA DATA2,DATA,READWRITE
RESULT DCD 0X0
```

```
END
```